



2018. Fall.

1. Let  $A \in M_{n \times n}(\mathbb{R})$  such that  $A^2 = 0$ . Prove

$$2 \cdot \text{rank } A \leq n.$$

Proof. Given  $x \in \mathbb{R}^n$ ,  $A(Ax) = A^2x = 0$ , so  $\text{Im } A \subseteq \ker A$ . So,  $\dim \text{Im } A \leq \dim \ker A$ .  
By the dimension theorem,

$$n = \text{nullity } A + \text{rank } A \geq 2 \text{rank } A \quad \square$$

2. Determine over  $\mathbb{R}$ ,

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

is diagonalizable.

Proof. The characteristic polynomial of  $A$  is

$$\det \begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3) + 2 \cdot (-1)(t-2) = (t-2)(t^2 - 3t + 2) = (t-1)(t-2)(t-2).$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to characteristic polynomial, so we need to check that  $(A-I)(A-2I) = 0$ .

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

So, the minimal polynomial of  $A$  is  $(t-1)(t-2)$ , it implies  $A$  is diagonalizable, because the minimal polynomial splits over  $\mathbb{R}$  and separable. Explicitly,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x = -2z \\ x+y+z=0 \end{cases} \Rightarrow \ker(A-I) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$$
$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x+z=0 \end{cases} \Rightarrow \ker(A-2I) = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

Therefore,  $A = UDU^{-1}$  where  $U = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$  and  $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ .

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3. Let  $R$  be a commutative ring and let  $x \in R$  such that  $x^{2018} = 0$ .  
prove that  $1+x$  is unit.

Proof. observe that

$$\begin{aligned} & (1+x) \cdot (-x^{2017} + x^{2016} - x^{2015} + \dots - x + 1) \\ &= 1 - x + x^2 + \dots + x^{2016} - x^{2017} \\ & \quad + x - x^2 + \dots - x^{2016} + x^{2017} - x^{2018} \\ &= 1, \quad \text{since } x^{2018} = 0. \quad \text{So, } 1 - x + \dots + x^{2016} - x^{2017} \text{ is inverse of } 1+x. \end{aligned}$$

4. prove that  $A_4$  does not have a subgroup of order 6.

Proof. since  $|S_4 : A_4| = 2$ ,  $|A_4| = 4 \cdot 3 = 12$ .

Suppose that  $A_4$  has a subgroup  $H$  of order 6.

Then,  $|A_4 : H| = 12/6 = 2$ , so  $gH = Hg$  for any  $g \in A_4$ , it means  $H \trianglelefteq A_4$  is normal.  
Consider the afforded Homomorphism of  $A_4$  acts  $A_4/H$  as left multiplication:

$$\mathcal{Q} : A_4 \rightarrow S_{A_4/H} : g \mapsto (x) \mapsto gxH$$

And, since every 3-cycle  $\sigma \in A_4$  satisfies

$$(\mathcal{Q}(\sigma))^3 = \mathcal{Q}(\sigma^3) = \mathcal{Q}(\text{id}) = \text{id}$$

and  $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$  so  $\mathcal{Q}(\sigma) = \text{id}$ . It implies all 3-cycles are contained in  $\ker \mathcal{Q}$ . Meanwhile, observe that  $(1\ 2\ 3), (1\ 2\ 4) \in A_4$  satisfies

$$(1\ 2\ 3)^{-1}(1\ 2\ 4)(1\ 2\ 3) = (1\ 3\ 2)(1\ 2\ 4)(1\ 2\ 3) = (1\ 4\ 3)$$

$(1\ 2\ 3)(1\ 4\ 3) = (1\ 4)(2\ 3)$ , ... in this way,  $A_4$  is generated by all 3-cycles in  $A_4$ , so  $\ker \mathcal{Q} = A_4$ . By the first isomorphism theorem,  $A_4/\ker \mathcal{Q} \cong \text{Im } \mathcal{Q}$  is trivial, so the action is trivial.

But, the left coset action is transitive, it cannot be trivial, contradiction.  $\square$

Similar Problems,

1. Show that  $A_5$  has no subgroup of order 15.

Proof.  $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$ .

Suppose that there is a subgroup  $H < A_5$  of order 15.

By the Sylow Theorem,  $\# \text{Syl}_3(H) \in \{1\}$  and  $\# \text{Syl}_5(H) \in \{1\}$ , that is,

there are order 3 subgroup  $P < H$  and order 5 subgroup  $Q < H$ ,

and both are normal subgroup of  $H$ . Since  $P \cong C_3$  and  $Q \cong C_5$  both are

cyclic, let  $x \in P$  and  $y \in Q$  such that  $P = \langle x \rangle$  and  $Q = \langle y \rangle$ .

Consider the afforded Homomorphism by  $H$  acts  $P$  by conjugation

$$\varphi: H \rightarrow \text{Sym}(P) \cong \text{Aut}(P) \quad (h \mapsto h x h^{-1})$$

Then,  $\text{Ker } \varphi = \{h \in H : h x h^{-1} = x \text{ for all } x \in P\} = C_H(P)$  and

$H / \text{Ker } \varphi \cong \text{Im } \varphi \leq \text{Aut}(P)$ . But, since  $\text{Aut}(P) \cong \mathbb{Z}/2\mathbb{Z}$ ,

and  $|H|$  has no factor 2, odd number, so  $|\text{Ker } \varphi| = |H|$ , that is,

$C_H(P) = H$ . Now,  $x$  and  $y$  commute, therefore  $|xy| = 15$ .

But,  $A_5$  cannot have order 15 element, because:

order of permutation in  $S_5$  is determined by the least common multiplier of disjoint cycle decompositions, but  $\text{lcm } 5 = 5$ ,

$\text{lcm}(4, 1) = 4$ ,  $\text{lcm}(3, 2) = 6$ ,  $\text{lcm}(2, 1, 1, 1) = 2$ , ...

the lcm cannot be 15, so  $H$  contradicts.

[Similar problem

Let  $G = S_3 \times C_2$ .

1. Find all normal subgroups of  $G$ .

2. Prove that  $G$  has no normal subgroup of order 4.

Proof. Define three maps:

$$\mathcal{Q}_1: G \rightarrow S_3: (\sigma, \alpha) \mapsto \sigma$$

$$\mathcal{Q}_2: G \rightarrow C_2: (\sigma, \alpha) \mapsto \alpha$$

$$\mathcal{Q}_3: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma \cdot \pi(\alpha) \quad \text{where } C_2 \stackrel{\pi}{\cong} \{\pm 1\}.$$

$$\mathcal{Q}_4: G \rightarrow \{\pm 1\} \times C_2: (\sigma, \alpha) \mapsto (\text{sgn } \sigma, \alpha).$$

$$\mathcal{Q}_5: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma$$

Then,  $\mathcal{Q}_1$  and  $\mathcal{Q}_2$  are Homomorphisms, being the Coordinatively Projection, and  $\mathcal{Q}_3$  is also Homomorphism, because  $\text{sgn}$ ,  $\pi$  are Homomorphism and  $\{\pm 1\}$  is abelian. Precisely,

$$\begin{aligned} \mathcal{Q}_3((\sigma_1, \alpha_1) \cdot (\sigma_2, \alpha_2)) &= \mathcal{Q}_3(\sigma_1 \sigma_2, \alpha_1 \alpha_2) = \text{sgn } \sigma_1 \text{sgn } \sigma_2 \pi(\alpha_1) \pi(\alpha_2) \\ &= \text{sgn } \sigma_1 \pi(\alpha_1) \text{sgn } \sigma_2 \pi(\alpha_2) = \mathcal{Q}_3(\sigma_1, \alpha_1) \cdot \mathcal{Q}_3(\sigma_2, \alpha_2). \end{aligned}$$

And,  $\mathcal{Q}_4$  is Homomorphism, being entrywise Homomorphism, and  $\mathcal{Q}_5$  is Homomorphism, being Composition of Projection and  $\text{sgn}$ .

Now, the kernel

$$\ker \mathcal{Q}_1 = \{\text{id}\} \times C_2, \quad \ker \mathcal{Q}_2 = S_3 \times \{x^0\}, \quad \ker \mathcal{Q}_3 = A_3 \times \{x^0\} \cup (S_3 \setminus A_3) \times \{x^1\}$$

$$\ker \mathcal{Q}_4 = A_3 \times \{x^0\}, \quad \ker \mathcal{Q}_5 = A_3 \times C_2.$$

are we desired.

Are these all possibilities?

Observe that: if  $N \subseteq G_1 \times G_2$  is normal, then the image of projections  $\pi_i: G_1 \times G_2 \rightarrow G_i: (g_1, g_2) \mapsto g_i$ ,  $\pi_i[N]$  is also normal of  $\pi_i[G_1 \times G_2] = G_i$ , because image of normal subgroup is normal in the Range, and the projections are onto. But, as we know, normal subgroup of  $S_3$  is only  $\{\text{id}\}$ ,  $A_3$ ,  $S_3$  and normal subgroup of  $\{\pm 1\}$  is  $\{1\}$  and  $\{\pm 1\}$ .

2018. Fall.

7. Consider the identity  $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$ .

- a). Determine the sequence  $(C_n) \subseteq \mathbb{R}$  and the range of  $x \in \mathbb{R}$  for the identity holds.  
b). Show that for each  $n \geq 0$ ,

$$\sum_{k=0}^n \frac{C_k}{(n-k)!} = 1.$$

Proof. a),

Notice that  $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$  for all  $x \in \mathbb{R}$  and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ for all } |x| < 1.$$

Therefore, the coefficient  $C_n$  is determined by the Cauchy product

$$C_n = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in  $\mathbb{R} \cap (-1, 1) = (-1, 1)$ .  $\perp 5m$ .

b). Since

$$\frac{1}{1-x} = e^{-x} \cdot \sum_{n=0}^{\infty} C_n x^n, \text{ so we can write for } x \in (-1, 1)$$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each  $n \geq 0$ , because these are infinitely differentiable, so the coefficients of power series are determined uniquely by the Taylor Theorem. Thus,

$$1 = \sum_{k=0}^n \frac{C_k}{(n-k)!} \text{ for all } n \geq 0. \perp 4m$$

Tzola, Spring

6. For each  $n \in \mathbb{N}$ , let

$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that  $(f_n)_{n \geq 1}$  does not converge uniformly.

b) For each fixed  $q > 1$ , Show that  $(f_n)$  converges uniformly on  $\{x \in \mathbb{R} \mid |x| \geq q\}$  and  $\{x \in \mathbb{R} \mid |x| \leq \frac{1}{q}\}$ .

a). Given  $x \in \mathbb{R}$ ,

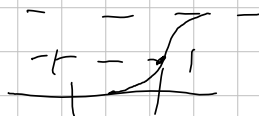
if  $|x| < 1$ , then  $f_n(x) \rightarrow 0$  as  $n \rightarrow \infty$ , since  $|x|^n \rightarrow 0$ .

if  $|x| = 1$ , then  $f_n(x) = \frac{1}{2}$  for all  $n \in \mathbb{N}$ , since  $|x|^n = 1$ .

if  $|x| > 1$ , then  $f_n(x) = \frac{1}{1/|x|^n + 1} \rightarrow 1$  as  $n \rightarrow \infty$ , since  $1/|x|^n \rightarrow 0$  as  $n \rightarrow \infty$ .

Therefore,  $f_n \rightarrow f$  pointwise where

$$f(x) = \begin{cases} 0 & |x| < 1 \\ \frac{1}{2} & |x| = 1 \\ 1 & |x| > 1 \end{cases}$$



But, for  $\epsilon = 1/4$ , and for all  $n \in \mathbb{N}$ ,

$$\sup_{0 \leq x < 1} |f_n(x) - f(x)| = \sup_{0 \leq x < 1} |f_n(x)| = \sup_{0 \leq x < 1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$

because given  $n \in \mathbb{N}$ , we can take  $x \in [0, 1)$  such that  $\frac{x^n}{1+x^n} \geq \frac{1}{4}$

Since  $\lim_{x \rightarrow 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$ , so,  $\sup_{x \in \mathbb{R}} |f_n - f| \geq \sup_{0 \leq x < 1} |f_n - f|$  implies

$f_n \rightarrow f$  but not uniformly on  $\mathbb{R}$ .

b). If  $|x| \geq q > 1$ , then  $f(x) = 1$ . So, enough to show that  $\sup |f_n(x) - f(x)| \rightarrow 0$ .

Since  $\sup_{|x| \geq q} |f_n(x) - f(x)| = \sup_{|x| \geq q} \left| \frac{1}{1+|x|^n} \right| \leq \frac{1}{1+q^n} \rightarrow 0$  as  $n \rightarrow \infty$ , so  $f_n \rightarrow f$  uniformly on  $|x| \geq q$ . Similarly, for  $|x| \leq \frac{1}{q}$ ,  $f(x) = 0$  and so

$$\sup_{|x| \leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x| \leq \frac{1}{q}} \left| \frac{|x|^n}{1+|x|^n} \right| \leq \frac{1}{q^n + 1} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

so proved all cases.  $\square$

2019, Spring.

17, Let  $S = \{(x, y, z) \in \mathbb{R}^3 \mid z = x \cdot f(\frac{y}{x}), x \neq 0\}$  where  $f: \mathbb{R} \rightarrow \mathbb{R}$  is differentiable. Show that the tangent plane at any point of  $S$  passes through the origin. Prove by the definition of  $S$ , given  $(x_0, y_0, z_0) \in S$ , a map

$X: \mathcal{O} \rightarrow S: (u, v) \mapsto (u, v, u \cdot f(\frac{v}{u}))$  where  $\mathcal{O}$  is an open set containing  $(x_0, y_0)$  is a parametrization of  $S$  at  $(x_0, y_0, z_0)$ . The differential of  $X$  at  $(x_0, y_0)$  is:

$$dX = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f(\frac{v}{u}) - \frac{v}{u} \cdot f'(\frac{v}{u}) & f'(\frac{v}{u}) \end{bmatrix} \bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of  $S$  at  $(x_0, y_0)$  is generated by

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} \cdot f'(\frac{y_0}{x_0}) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ f'(\frac{y_0}{x_0}) \end{pmatrix} \right\}.$$

Now, for  $t_1, t_2 \in \mathbb{R}$ , the coordinate of a point of the tangent plane can be described by

$$(x_0, y_0, x_0 \cdot f(\frac{y_0}{x_0})) + (t_1, 0, t_1 \cdot f(\frac{y_0}{x_0}) - t_1 \cdot \frac{y_0}{x_0} f'(\frac{y_0}{x_0})) \\ + (0, t_2, t_2 \cdot f'(\frac{y_0}{x_0}))$$

is zero when  $t_1 = -x_0$  and  $t_2 = -y_0$ , so proved  $\square$



201a, Fall.

8. Let  $(b_n)_{n \in \mathbb{N}}$  be a decreasing sequence of positive real numbers with  $b_n \rightarrow 0$ . Let  $(a_n)_{n \in \mathbb{N}}$  be a sequence of complex numbers such that  $(\sum_{n=1}^N a_n)_{N \in \mathbb{N}}$  is a bounded sequence.

a). Show that  $\sum_{n=1}^{\infty} a_n b_n$  converges.

b). Show  $\sum_{n=2}^{\infty} \frac{\sin(n\pi x)}{\log n}$  converges for all  $x \in [-\pi, \pi]$ .

Proof. Denote  $A_n = \sum_{k=1}^n a_k$  and  $A_0 = 0$ . And let  $M \geq 0$  be a bound for  $\sum a_n$ .

$$\begin{aligned} \sum_{n=1}^N a_n b_n &= \sum_{n=1}^N [A_n - A_{n+1}] \cdot b_n = \sum_{n=1}^N A_n b_n - \sum_{n=1}^N A_{n+1} b_n \left( \sum_{n=1}^N A_n b_n = \sum_{n=0}^{N-1} A_n b_{n+1} \right) \\ &= \sum_{n=1}^N A_n b_n - \left( \sum_{n=1}^N A_n b_{n+1} + A_0 b_1 - A_N b_{N+1} \right) \\ &= \sum_{n=1}^N A_n (b_n - b_{n+1}) + A_N b_{N+1}. \end{aligned}$$

$$\sum_{k=n}^m a_k b_k = \sum_{k=n}^m A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_n b_n$$

$$\begin{aligned} \left| \sum_{k=n}^m a_k b_k \right| &\leq \sum_{k=n}^m |A_k| \cdot |b_k - b_{k+1}| + |A_m| |b_{m+1}| + |A_n| |b_n| \\ &\leq M \cdot \left( \sum_{k=n}^m |b_k - b_{k+1}| + |b_{m+1}| + |b_n| \right) \\ &= M \cdot \left( \sum_{k=n}^m (b_k - b_{k+1}) + b_{m+1} + b_n \right) \end{aligned}$$

$$= M \cdot 2 b_n \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ so proved.}$$

b). Since  $1/\log n$  is decreasing and positive real numbers, so by a), enough to show that  $|\sum \sin(n\pi x)|$  is bounded for all  $x \in [-\pi, \pi]$ .

observe that:  $2 \sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

$$\text{Therefore, } 2 \sin x \cdot \sum_{n=1}^N \sin(n\pi x) = \sum_{n=1}^N [\cos((n-1)x) - \cos((n+1)x)] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos(N+1)x| \leq 4$$

so for  $x \in [-\pi, \pi] \setminus \{0, \pm\pi\}$ ,  $\left| \sum_{n=2}^N \sin n\pi x \right| \leq \frac{2}{|\sin x|}$  for any  $N \in \mathbb{N}$ , bounded.  
for  $x = 0$  or  $\pm\pi$ , then the sum is just zero.  $\square$

2019, Fall,

1. Let  $\varphi: [0,1] \rightarrow \mathbb{R}$  be a continuous map. Evaluate

$$\lim_{\varepsilon \rightarrow 0} \int_0^1 \left[ \frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of  $y'' - 3y' + 2y = 0$  is  $r^2 - 3r + 2 = (r-1)(r-2)$ , therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^t + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution  $y_p(t)$ . First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$\begin{aligned} y_p(t) &= \int_{x_0}^x \begin{vmatrix} e^t & e^{2t} \\ e^{2t} & e^{2t} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}} dt \\ &= \int_{x_0}^x \frac{e^{t+2t} - e^{2t+t}}{e^{3t} + e^{2t}} dt \\ &= e^{2x} \int_{x_0}^x \frac{1}{e^{2t} + e^t} dt - e^x \int_{x_0}^x \frac{1}{e^t + 1} dt \end{aligned}$$

Meanwhile, since  $\int \frac{1}{e^t + 1} dt = \int \frac{e^{-t}}{1 + e^{-t}} dt = -\log(1 + e^{-x})$

and  $\int \frac{1}{e^{2t} + e^t} dt = \int \frac{e^{-t}}{e^t + 1} dt = \int \frac{-u}{\frac{1}{u} + 1} \cdot \frac{1}{u} du = \int \frac{-u}{1+u} du$

$$= \int -1 + \frac{1}{1+u} du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$\begin{aligned} y_p + C_1 e_1 + C_2 e_2 &= \left[ e^x + e^{2x} \log(1+e^{-x}) + e^x \cdot \log(1+e^x) \right] \\ &+ C_1 e^x + C_2 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R}, \end{aligned}$$

12/20, Fall.

1. Let  $p$  be a prime. Find the number of subgroups  $H$  of  $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ ,  $|H| = p^2$ .  
Proof. By the fundamental theorem of finitely generated abelian group, either  $|H| \cong \mathbb{Z}_{p^2}$  or  $\mathbb{Z}_p \times \mathbb{Z}_p$ .

If  $|H| \cong \mathbb{Z}_p \times \mathbb{Z}_p$ , then for all  $x \in H$ ,  $p \cdot x = 0$ . Therefore,  
 $H \subseteq \{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\}$ .

Meanwhile,  $pa = 0$  and  $pb = 0$  if and only if  $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$  and  $b \in \langle p^2 \rangle \subseteq \mathbb{Z}_{p^3}$ ,  
so  $\{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$ .

Since  $\langle p \rangle \times \langle p^2 \rangle$  has already order  $p^2$ , so  $H = \langle p \rangle \times \langle p^2 \rangle$ .

If  $|H| \cong \mathbb{Z}_{p^2}$ , that is,  $H$  is generated by order  $p^2$  element,  
Since  $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$  has order  $p^2 \iff p^2 \cdot x = 0$  and  $px \neq 0$ .

Every element  $a \in \mathbb{Z}_{p^2}$  satisfies  $p^2 \cdot a = 0$ , and  $b \in \mathbb{Z}_{p^3}$  satisfies  $p^2 \cdot b = 0$   
if and only if  $b \in p\mathbb{Z}_{p^3}$  and  $\mathbb{Z}_{p^3}$  has number of  $p^2$  of  $p$ -multiples,  
so there are  $p^2 \cdot p^2 = p^4$  elements satisfying  $p^2 \cdot x = 0$ .

And,  $px = 0$  if and only if  $a \in p\mathbb{Z}_{p^2}$  and  $b \in p^2\mathbb{Z}_{p^3}$ , where  $x = (a, b)$ .  
So, there are  $p^2$  elements satisfying  $px = 0$ .

Consequently, there are  $p^4 - p^2$  elements  $x$  such that  $|H| = \langle x \rangle$ .

Moreover, the cyclic group of order  $p^2$  has  $\phi(p^2) = p^2 - p$  generators,

so the possible  $|H| \cong \mathbb{Z}_{p^2}$  are

$$(p^4 - p^2) / (p^2 - p)$$

numbers.

Finally, the number of  $|H|$  is  $1 + (p^4 - p^2) / (p^2 - p)$ .

Note: for a prime  $p$ ,  $\phi(p^n) = p^n - p^{n-1}$ .

2020 Fall.

2. For each  $n \in \mathbb{N}$ , let  $a_n = \cos(2\pi/n)$ .

a). Prove that  $a_n$  is algebraic over  $\mathbb{Q}$ .

b). Calculate  $\deg_{\mathbb{Q}}(a_5)$ .

Proof. a). Let  $\zeta = \exp(2\pi i/n)$ . Then,  $\zeta^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$  gives that  $a_n = \cos(2\pi/n) = (\zeta + \zeta^{-1})/2$ .

Meanwhile, since  $\zeta^n = \exp 2\pi i = 1$ , so  $\zeta^n - 1$  gives  $\zeta$  is algebraic over  $\mathbb{Q}$ .

Therefore,  $\zeta^{-1}$  is also algebraic, thus  $a_n = \frac{\zeta + \zeta^{-1}}{2}$  is algebraic over  $\mathbb{Q}$ .

b). Let  $w = \exp(2\pi i/5)$ . Since  $w$  is the primitive 5th root of unity,  $w^4 + w^3 + w^2 + w + 1 = 0$ . And, dividing by  $w^2$  then  $2a_5 = w + w^{-1}$  implies

$$w^4 + w^3 + w^2 + w + 1 = 0 \Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0$$

$$\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0$$

$$\Rightarrow 4a_5^2 + 2a_5 - 1 = 0.$$

Since  $4x^2 + 2x - 1$  is irreducible over  $\mathbb{Q}$  by the discriminant  $2^2 + 16 = 20$  but  $\sqrt{20} \notin \mathbb{Q}$ , so  $\deg_{\mathbb{Q}}(a_5) = 2$ .  $\square$

Fall.

a) Notice that De Moivre's formula gives  $e^{i \cdot n z} = (\cos n z + i \cdot \sin n z) = (\cos z + i \cdot \sin z)^n$

Therefore

$$\begin{aligned} \cos n \cdot \frac{2\pi}{n} = 1 &= \operatorname{Re} \left( \cos \frac{2\pi}{n} + i \cdot \sin \frac{2\pi}{n} \right)^n \\ &= \operatorname{Re} \left( \sum_{k=0}^n \binom{n}{k} \left( \cos \frac{2\pi}{n} \right)^k \cdot \left( i \cdot \sin \frac{2\pi}{n} \right)^{n-k} \cdot \binom{n}{k} \right) \end{aligned}$$

Meanwhile,  $\sin^2(\frac{2\pi}{n}) = 1 - \cos^2(\frac{2\pi}{n})$  and

the real part of  $\sum_{k=0}^n \binom{n}{k} \cdot \left( \cos \frac{2\pi}{n} \right)^k \cdot \left( i \cdot \sin \frac{2\pi}{n} \right)^{n-k}$  yield only  $n-k$  is even,

since  $i^{n-k}$  is real  $\Leftrightarrow n-k$ , so  $\operatorname{Re} \left( \sum_{k=0}^n \binom{n}{k} \cdot \left( \cos \frac{2\pi}{n} \right)^k \cdot \left( i \cdot \sin \frac{2\pi}{n} \right)^{n-k} \right) - 1 = 0$  is evaluated  $\cos(\frac{2\pi}{n})$  for some polynomial over  $\mathbb{Q}$ , so  $a_n$  is algebraic over  $\mathbb{Q}$ .

b). From above argument,

$$\begin{aligned} 1 &= \underbrace{1 \cdot a_5^0}_{k=0} + \underbrace{5 \cdot a_5^1 \cdot (1 - a_5^2)}_{k=1} - \underbrace{10 \cdot a_5^3 \cdot (1 - a_5^2)^2}_{k=3} + \underbrace{a_5^5}_{k=5} \\ &= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5 \\ &= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7. \end{aligned}$$

$$\text{So, } 10 a_5^7 - 11 a_5^5 + 15 a_5^3 - 5 a_5 + 1 = 0.$$

2020, Fall

3. Let  $R$  be a field. Show that  $(x^3 - y^5) \in R[x, y]$  is prime ideal.

Proof. Since  $R[x, y]$  is U.F.D., every irreducible element is prime.

Thus, enough to show that  $x^3 - y^5$  is irreducible.

Notice that:  $R[x, y] = R[y][x]$  by definition, so  $x^3 - y^5 \in R[y][x]$ .

Since  $\gcd(1, y^5) = 1$ ,  $x^3 - y^5$  is primitive, by the Gauss Lemma,

$x^3 - y^5$  is irreducible over  $R[y] \Leftrightarrow x^3 - y^5$  is irreducible over  $R(y)$ .

But, since  $x^3 - y^5$  has degree 3,  $x^3 - y^5$  is reducible only if it has a linear factor, in other say,  $x^3 - y^5$  has a root in  $R(y)$ .

Suppose that for some  $f(y)/g(y) \in R(y)$ ,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then,  $f(y)^3 = y^5 g(y)^3$ , But  $\deg f(y)^3 \equiv 0 \pmod{3}$ , and

$$\deg y^5 g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \pmod{3}.$$

Therefore, there is no such a root, so  $x^3 - y^5$  is irreducible over  $R(y)$ ,  
so it is irreducible over  $R[x, y]$ .

Note:

$$K[x, y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test.}}$$

Modified

Prove that  $\gcd(m,n)=1$  implies  $x^m - y^n \in F[x,y]$  is irreducible.

2020. Fall.

4. Let  $W \subseteq V = \mathbb{R}^n$  be a subspace. Show that there exists a homogeneous system of linear equations whose solution space is  $W$ .

Proof. Let  $\mathcal{B} = \{\alpha_1, \dots, \alpha_k\}$  be a basis for  $W$ .

Then, we can extend the basis  $\mathcal{B}$  to  $\overline{\mathcal{B}} = \mathcal{B} \cup \{\alpha_{k+1}, \dots, \alpha_n\}$  for  $V$ .

Now, consider the map:

$$f_{\mathcal{B}}: V \rightarrow \mathbb{R}^k; \sum c_i \alpha_i \mapsto C_{\mathcal{B}}$$

Then,  $W = \ker f_{k+1} \cap \dots \cap \ker f_n$ , so for the standard basis  $\mathcal{S}$  for  $V$ ,

$$A = \begin{pmatrix} [f_{k+1}]_{\mathcal{S}} \\ \vdots \\ [f_n]_{\mathcal{S}} \end{pmatrix} \text{ satisfies } \ker A = W, \text{ so}$$

([ $f$ ] $_{\mathcal{S}}$  is row vector representation).

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ is the system we desired.}$$



2021. Spring.

1. Show that no group of 2020 is simple.

Proof  $2020 = 2^2 \cdot 5 \cdot 101$ . Denote  $n_p$  as the number of Sylow  $p$ -subgroups.

By the Sylow theorem,  $n_{101} \equiv 1 \pmod{101}$  and  $n_{101} \mid 20$ , so  $n_{101}$  must be 1. Therefore, the Sylow 101-subgroup is normal so proved.

Without using Sylow theorem.

By the Cauchy's theorem,  $G$  has cyclic subgroup  $H \leq G$  of order 101.

Denote  $A = \{g \mid g^{-1} \notin G\}$ . Then,  $A$  is orbit of  $H$  from the action

$$G \times \{S \leq G\} \rightarrow \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of  $H$  is  $G_H = N_G(H)$ , we obtain

$$\#A = |G : N_G(H)|.$$

And since  $H \leq N_G(H)$ ,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \#A \leq 20.$$

Now, consider the action

$$H \times A \rightarrow A : (h, K) \mapsto hKh^{-1}.$$

Given  $K \in A$ , the  $H$ -orbit of this action is  $\text{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$  and

$$\#\text{Orb}_H(K) = |H : \text{Stab}_H(K)|, \text{Stab}_H(K) \leq H \Rightarrow |\text{Stab}_H(K)| \in \{1, 101\}.$$

So,  $\#\text{Orb}_H(K) \in \{1, 101\}$  and  $\text{Orb}_H(K) \leq A$ , so  $\#\text{Orb}_H(K) = 1$ .

Therefore, every element in  $H$  normalizes  $K \in A$ , that is,  $H \leq N_G(K)$ .

Now, for some  $K \in A$ , if  $H \neq K$  then  $HK \leq G$  and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since  $101^2 \nmid 2020$ , so  $A = \{1\}$ . That is,  $H \leq G$   $\square$

2021. Spring

7. Let  $f$  be a polynomial given by

$$f(z) = a_0 + a_1 z + \dots + a_d z^d \quad \text{over } \mathbb{C}, \quad a_d \neq 0$$

8m.

Evaluate the value  $\lim_{R \rightarrow \infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz$

Proof. By the fundamental theorem of Algebra,  $f$  has  $d$ -roots  $z_1, \dots, z_d \in \mathbb{C}$ .

That is,  $f(z) = a_d (z - z_1) \dots (z - z_d)$

Moreover,  $\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \dots + \frac{1}{z - z_d}$ , just calculation.

Therefore, for  $R > \max\{|z_1|, \dots, |z_d|\}$ , by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz = \sum_{n=1}^d \operatorname{Res}\left(\frac{z}{z - z_n}, z_n\right) = \sum_{n=1}^d \lim_{z \rightarrow z_n} z = z_1 + \dots + z_d$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \dots + z_d) = 2\pi i \cdot \left(\frac{-a_{d-1}}{a_d}\right),$$

□

2022 Spring.

8. Define  $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$

Then,  $f$  is differentiable on  $\mathbb{R}$ .

Proof. For each  $m \in \mathbb{N}$ , define

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

At first,  $f_m(x) \rightarrow f(x)$  as  $m \rightarrow \infty$ , because that

$$\sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1 \text{ as } m \rightarrow \infty,$$

so the increasing bounded sequence  $f_m(x)$  converges. Now,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

if  $|x| \leq 1$ , then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{2}{n^4} \text{ and } \sum \frac{1}{n^4} < \infty,$$

if  $|x| > 1$ , then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq 2 \cdot \frac{|x|}{x^2 + n^2} \cdot \frac{1}{x^2 + n^2} \leq 2 \cdot \frac{1}{n^2} \text{ and } \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives  $f'_m$  converges uniformly.

Now,  $f_m$  converges at some point and  $f'_m$  converges uniformly on  $\mathbb{R}$ , the limit function  $f$  is differentiable.  $\square$

First, for any  $x \in \mathbb{R}$ ,  $|x^2 + n^2| \geq n^2$ , so

$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2 + n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1,$$

as  $m \rightarrow \infty$ . Therefore, given  $\varepsilon_0$ , for some  $N \in \mathbb{N}$ ,  $m > n \geq N$  implies

$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^m \frac{1}{x^2 + k^2} \right| \leq \sup_{x \in \mathbb{R}} \sum_{k=n}^m \left| \frac{1}{x^2 + k^2} \right| < \sum_{k=n}^m \frac{1}{k^2} < \varepsilon.$$

Now, define for each  $m \in \mathbb{N}$ ,

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

We already shown  $f_m \rightarrow f$  uniformly. And,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

2022. Fall.

1. Let  $G$  be a group and  $H$  and  $K$  are normal subgroup of  $G$ .

If  $H \cap K = \{e\}$ , then  $H/K \cong H \times K$ .

Proof. Define a map

$$\varphi: H/K \rightarrow H/K: (h, k) \mapsto hk.$$

It is well-defined by the definition of  $H/K$ , and it is homomorphism; given  $h_1, h_2 \in H$  and  $k_1, k_2 \in K$ ,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 k_1 k_2$$

meanwhile, given  $h \in H$  and  $k \in K$ ,  $hkh^{-1}k^{-1} \in K$  and  $\in H$ , because  $hkh^{-1} \in K$  and  $kh^{-1}k^{-1} \in H$ , by the normality. But,  $H \cap K = \{e\}$  implies  $hkh^{-1}k^{-1} = e$ , so  $hk = kh$ , these commute so,

$$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1)) \varphi((h_2, k_2)), \text{ so } \varphi \text{ preserves the operation.}$$

And,  $\varphi[H/K] = H/K$  by just definition,  $\varphi$  is onto. Now, the first isomorphism theorem gives that  $H/K / \ker \varphi \cong H/K$ .

$$\text{And, } \ker \varphi = \{(h, k) \in H \times K \mid hk = e\} = \{e\},$$

so  $H/K \cong H/K / \{e\}$ , it proves the assertion.  $\square$

2022 Fall.

8. For continuous functions  $g: [0,1] \rightarrow \mathbb{R}$  and  $h: [-1,1] \rightarrow \mathbb{R}$ , define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \int_0^1 h(x-y)g(y) dy$$

For each  $n \in \mathbb{N}$ , define

$$f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sum_{i=0}^{n-1} h(x - \frac{i}{n}) \cdot g(\frac{i}{n})$$

Prove that  $f_n \rightarrow f$  uniformly on  $[0,1]$

Proof. Define  $l: [0,1] \times [0,1]; (x,y) \mapsto h(x-y) \cdot g(y)$ .

Since  $h$  and  $g$  both are continuous, therefore  $l$  is continuous and since  $[0,1]^2$  is compact set, so  $l$  is continuous uniformly.

Given  $\epsilon > 0$ , there exists  $\delta > 0$  such that for any  $(x_1, y_1), (x_2, y_2) \in [0,1]^2$ ,

$$\|(x_1, y_1) - (x_2, y_2)\| < \delta \Rightarrow \|l(x_1, y_1) - l(x_2, y_2)\| < \epsilon.$$

Now, given  $x \in [0,1]$ , and  $n > \frac{1}{\delta}$ , ( $n > \frac{1}{\delta}$  gives  $\frac{1}{n} < \delta$ )

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \int_0^1 l(x, y) dy - f_n(x) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \frac{1}{n} \cdot \sum_{i=0}^{n-1} l(x, \frac{i}{n}) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \sum_{i=1}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, \frac{i}{n}) dy \right| \\ &\leq \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} |l(x, y) - l(x, \frac{i}{n})| dy \\ &< \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \epsilon dy = \epsilon. \end{aligned}$$

Therefore,  $n > \frac{1}{\delta} \Rightarrow \sup_{x \in [0,1]} |f_n(x) - f(x)| < \epsilon$ , So, proved.  $\square$

12022. Fall.

18. Let  $S$  be the subset of  $\mathbb{R}^3$  given by  $\{(x, y, z) \mid z^2 = x^2 + y^2, z \geq 0\}$ .

Show that  $S$  is not regular surface.

Proof. Suppose that  $S$  is a regular surface. Then, for  $(0, 0, 0) \in S$ , there exists a neighbor  $V$  containing  $(0, 0, 0)$  such that  $V \cap S$  is a graph of

$$z = f(x, y) \quad \text{or} \quad y = g(x, z) \quad \text{or} \quad z = h(y, z)$$

where  $f, g, h$  are differentiable. Since projection of  $S$  into  $xz$ ,  $yz$ -plane are not 1-1,  $g, h$  cannot be such a map.

Meanwhile,  $f(x, y) = \sqrt{x^2 + y^2}$  at locally  $(0, 0, 0)$ , but  $f$  is not differentiable at  $(x, y) = (0, 0)$ , so it cannot be possible. Therefore  $S$  is not regular.

(More specifically,  $(0, \pm \epsilon, \sqrt{1 - \epsilon^2}) \in S$  for any  $\epsilon > 0$ , so there is no 1-1  $yz$ -projection, similarly  $xz$ .)

2023. Spring

4. Let  $V = \{\text{id}, (12)(34), (13)(24), (14)(23)\} \subseteq S_4$ .

a). Show that  $V$  is normal subgroup.

b).  $S_4/V \cong S_3$ ,

2023, Spring.

7. Let  $f: \mathbb{C} \rightarrow \mathbb{C}$  be an entire satisfying 10m

$$U^2(z) - 2 \cdot V^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that  $f$  is constant.

Proof. Observe that ~~differentiating gives~~ that

$$2U U_x - 2V V_x = 0 \Rightarrow \cancel{U U_x} = V V_x = -V \cdot U_y$$

Observe that:  $f(z) = 0 \Leftrightarrow U(z) = V(z) = 0$

But,  $U^2 - 2V^2 = 1$ , so  $f(z) \neq 0$  for all  $z \in \mathbb{C}$ . Now,

$$\frac{1}{f(z)} = \frac{1}{U^2(z) + V^2(z)} = \frac{1}{1 + 3V^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function  $\frac{1}{f^2}$  is constant, so  $f$  is also constant function.  $\square$

$$\text{Or, } |f^2| = U^2 + V^2 = 1 + 3V^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3V^2}} \leq 1.$$



2024, Spring.

1. Let  $A, B \in M_{m \times n}(F)$ . Show that  $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$ . 5m

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given  $\alpha \in \text{Im}(A+B)$ ,  $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$  for some  $\beta \in F^n$ , therefore  $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$ . Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that  $P$  and  $Q$  are positive definite such that  $P^2 = Q^2$ . Show  $P = Q$ .

Proof. Since  $P$  is Hermitian, there exists an orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  such that  $P\alpha_i = C_i\alpha_i$  for some  $C_i > 0$ , for each  $i$ .

Let  $1 \leq i \leq n$  be fixed. Then, by the assumption,

5m

$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2 I)\alpha_i = (Q - C_i I)(Q + C_i I)\alpha_i = 0.$$

Since  $Q$  is positive definite,  $(Q - C_i I)\alpha_i = 0$ , because: if it is not zero, then  $(Q - C_i I)(Q + C_i I)\alpha_i = (Q + C_i I)(Q - C_i I)\alpha_i = 0$ ,

so  $-C_i$  is negative eigenvalue of  $Q$ , it contradicts to  $Q$  is positive definite. Therefore, for each  $1 \leq i \leq n$ ,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since  $\mathcal{B}$  is basis, therefore  $P = Q$ .  $\square$

2024, Spring

3. Let  $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$  where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists  $T$ -invariant subspace  $V \subseteq \mathbb{R}^5$  s.t.  $\dim V = 4$ ?

b). "

s.t.  $\dim V = 2$ ,  $T|_V$  is diagonalizable?

c). "

$\dim V = 3$ , "

Proof.

a). Let  $V = \langle e_1, e_2, e_3, e_4 \rangle$ . Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in  $V$ , so  $T[V] \subseteq V$ . (clearly  $\dim V = 4$ )

b). Let  $V = \langle e_1, e_3 \rangle$ . Then  $Ae_1 = e_1$ ,  $Ae_3 = e_3$  are contained in  $V$ , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of  $T$  is  $(t-1)^5$ .

If such a subspace  $V$  exists, then the characteristic polynomial of  $T|_V$  is

$(t-1)^3$ . Since  $T|_V$  is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of  $T|_V$  must be  $(t-1)$ .

That is,  $T|_V = I_V$  but it implies that  $\ker(T-I)$  has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

2024, Spring.

5. Let  $K$  be a field, and  $L_1, L_2 \subseteq \overline{K}$  be finite extensions of  $K$ . 41

a). Prove  $L = L_1 L_2$  is finite extension.

b). If  $L_1$  and  $L_2$  are Galois over  $K$ , then  $L/K$  is Galois.

a). Since  $L_1$  and  $L_2$  both are finite, denote

$$L_1 = K(\alpha_1, \dots, \alpha_n) \quad \text{and} \quad L_2 = K(\beta_1, \dots, \beta_r) \quad \text{where } \alpha_i, \beta_j \in \overline{K} \text{ are algebraic over } K.$$

$$\text{Now, } L = L_1 L_2 = \bigcap_{\substack{E \subseteq \overline{K} \\ L_1, L_2 \subseteq E}} E = K(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_r)$$

is finitely generated by algebraic elements  $\alpha_i, \beta_j$ , so  $L$  is finite. 14m.

b). Since  $L_1$  and  $L_2$  are splitting field of  $K$ , let  $f_1$  and  $f_2$  be polynomials over  $K$  such that  $L_i$  is the splitting field of  $f_i$ ,  $i=1, 2$ . It is possible, because  $L_1, L_2$  both are finite extension.

Now let  $E$  be the splitting field of  $f_1, f_2$ . Then,  $E \subseteq L_1 L_2$ , because  $E$  is the smallest field containing all roots of  $f_1, f_2$ ,  $L_1 L_2 \subseteq E$  because  $E$  contains all roots of  $f_1$  and  $f_2$ . So,  $L = E$ ,  $E$ , therefore  $L$  is splitting field over  $K$ . 17m.

Since  $L_1$  and  $L_2$  both are Galois, for some separable  $f_1$  and  $f_2$  over  $K$ ,  $L_1$  and  $L_2$  are splitting field of  $f_1$  and  $f_2$  respectively.

$L_1 L_2$  is  $f_1, f_2$  splitting field  $\Rightarrow OK$

즉  $f_1, f_2$  is separable 이란 보장 X.

중복 linear factor 소거  $\Rightarrow f_1, f_2$ 가 irreducible over  $K$  인가?

Revised in the Algebra/Field Theory/

1/22/24 spring.

8. Let  $C_b(\mathbb{R})$  be the set of all real-valued bounded continuous functions on  $\mathbb{R}$ . For two functions  $f, g \in C_b(\mathbb{R})$ , define

$$d(f, g) := \sup_{x \in \mathbb{R}} |f(x) - g(x)|.$$

Let  $\{f_n\}_{n=1}^{\infty}$  be a sequence in  $C_b(\mathbb{R})$  such that:

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$  such that  $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$ .

a). Prove that there exists  $f \in C_b(\mathbb{R})$  such that  $\lim_{n \rightarrow \infty} d(f, f_n) = 0$

b). Prove that  $\lim_{n \rightarrow \infty} \left( \sup_{x \in \mathbb{R}} |f_n(x)| \right) = \sup_{x \in \mathbb{R}} |f(x)|$ .

[Proof. For each  $x \in \mathbb{R}$ , a sequence  $\{f_n(x)\}$  in  $\mathbb{R}$  is Cauchy sequence by hypothesis. Since  $\mathbb{R}$  is complete,  $\lim_{n \rightarrow \infty} f_n(x)$  exists, put this  $f(x) \in \mathbb{R}$ .

a).

Since  $x \in \mathbb{R}$  was arbitrary,  $f(x)$  is well-defined on  $\mathbb{R}$ , but still not necessarily  $f \in C_b(\mathbb{R})$ , and  $d(f, f_n) \rightarrow 0$ .

Given  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that  $m > n \geq N$  implies that  $\sup_{x \in \mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$ .

So, for any  $x \in \mathbb{R}$ ,  $|f_m(x) - f_n(x)| \leq \varepsilon$ . Letting  $m \rightarrow \infty$ , we obtain  $|f(x) - f_n(x)| \leq \varepsilon$ .

Since  $x$  was arbitrary,  $\sup_{x \in \mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$ , for  $n \geq N$ .

Therefore, the triangle inequality gives  $|f(x)| \leq |f_n(x)| + \varepsilon$  for all  $x \in \mathbb{R}$

and since  $f_n$  was bounded,  $f$  is bounded. And, the argument above

proves  $\sup_{x \in \mathbb{R}} |f - f_n| \rightarrow 0$  as  $n \rightarrow \infty$ , so since  $f_n$  is continuous and

$f_n \rightarrow f$  uniformly, so  $f$  is also continuous, so  $f \in C_b(\mathbb{R})$  and a) is proved.

b). Since  $\left| \sup_{x \in \mathbb{R}} |f_n(x)| - \sup_{x \in \mathbb{R}} |f(x)| \right| \leq \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| \rightarrow 0$  as  $n \rightarrow \infty$

Prove directly.

2025 Spring.

13. Let  $D^2$  be a closed unit disc in  $\mathbb{R}^2$  centered at the origin.

a). Show that the quotient space  $\mathbb{R}^2/D^2$  is homeomorphic to  $\mathbb{R}^2$ .

b). Let  $U = \text{int } D$ . Determine  $\mathbb{R}^2/U \cong \mathbb{R}^2$ .

Proof. Let  $\sim$  be a relation defined on  $\mathbb{R}^2$  as:

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \{(x_1, y_1), (x_2, y_2)\} \subseteq D^2.$$

Then, it is clearly an equivalence relation, and  $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$ , by definition.

a). Define a map

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2; (x, y) \mapsto$$

Then, given  $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ ,

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow$$

2025. Spring.

14. For  $a, b, c > 0$ , Consider the ellipsoid

$$E_{a,b,c} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

- a). If  $a=b$ , then  $E_{a,a,c}$  is a surface of revolution (Prüferl.)
- b) Give a parametrization of  $E_{a,a,c}$ . Compute the first fundamental form of this parametrization and the Christoffel symbol.
- c). Show that the Curve  $E_{a,b,c} \cap \{x=0\}$  is the image of a geodesic.

a). Let

$$X: (0, 2\pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3; (u, v) \mapsto (a \cdot \cos v \cdot \cos u, a \cdot \cos v \cdot \sin u, c \cdot \sin v).$$

2025, Fall

1. a). Show that  $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$  is linear functional.

8m

b). Find a dual basis corresponding to  $\{1, t, \dots, t^n\}$  of  $P_n(\mathbb{R})$ .

Proof. Let  $f(x) = \sum_{i=0}^n a_i x^i$  and  $g(x) = \sum_{i=0}^n b_i x^i$  and  $C \in \mathbb{R}$  be given.

Then,  $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each  $k=0, 1, \dots, n$ ,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in  $\mathcal{B}^*$  are all linear functional, so the  $\mathcal{B}^*$  is dual basis.

2025. Fall

2. Let  $A \in M_{\text{max}}(1\mathbb{R})$ .

15m:  $A^n$ 의 rank,  $n=2$ 일 때부터 시작

a). For  $n \in \mathbb{N}$ ,  $\text{rank } A^n \leq \text{rank } A$ .

b). Let  $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$ .

Show that for  $n \geq 2$ ,

$$\text{rank } A^n = \text{rank } A \text{ implies } \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given  $x \in \text{Im } A^n$ ,  $x = A^n y$  for some  $y \in \mathbb{R}^m$ .

So,  $x = A^n y = A^{n-1} A y \in \text{Im } A$ , for  $n \geq 2$ . Thus  $\text{Im } A^n \subseteq \text{Im } A$ , it follows  $\text{rank } A^n \leq \text{rank } A$ . If  $n=1$ , clear.

b). Given  $x \in \ker A$ ,  $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$ , so  $x \in \ker A^n$ , so  $\ker A \subseteq \ker A^n$ . Since  $\text{rank } A^n = \text{rank } A$ , the dimension theorem gives  $\text{nullity } A^n = \text{nullity } A$ .

Therefore, since  $\mathbb{R}^m$  is finite dimensional, so is  $\ker A^n$ , thus  $\ker A = \ker A^n$ .

Now, let  $x \in \ker L_A \cap \text{Im } L_A$  be given. Then,  $Ax = 0$  and  $x = Ay$  for some  $y \in \mathbb{R}^m$ .

Then,  $A^{n-1} x = A^n y = 0$ , because  $Ax = 0$ , so  $A^{n-2} Ax = 0$ , and this implies  $y \in \ker A^n$ .

Since  $\ker A^n = \ker A$ ,  $Ay = 0$ , so  $x = Ay = 0$ , so proved.  $\square$



2025. Fall

3. Find all  $a, b, c \in \mathbb{R}$  such that

6m

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of  $A$  is regardless of  $a, b, c$ ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the  $A$  is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore,  $a=0$ ,  $b \in \mathbb{R}$ ,  $c \in \mathbb{R}$  are all possible  $(a, b, c)$ .

2025. Fall,

4. Let  $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$  be the set of all pairs  $(A, B)$  such that  $\det A = \pm \det B$ .

20m

Show that  $N$  is normal and determine the factor group  $(G \times G)/N$ .

Sol). Let  $(A, B) \in N$  and  $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$  be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

$$\text{So, since } (A, B) \in N, \det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$$

therefore  $(C, D)(A, B)(C, D)^{-1} \in N$ . And, to show  $N$  is subgroup,

first,  $(I, I) \in N$  trivially, so  $N \neq \emptyset$ . Let  $(A, B), (C, D) \in N$  be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

$$\text{satisfies } \det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}.$$

therefore, by the subgroup criterion,  $N$  is subgroup, and normal.  $\perp$  hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B^{-1})^2$$

Then,  $\ker \mathcal{Q} = N$  and  $\mathcal{Q}$  is homomorphism, onto  $(0, \infty)$ , because

the det preserves multiplication, so is  $\mathcal{Q}$ , and onto is given by: fix  $B = I$ , and  $A = \text{diag}(r, 1, \dots, 1)$  where  $r > 0$  is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \xrightarrow{\perp} \text{hom.} \quad \mathcal{Q}$$

Comment:  $\mathcal{Q}$  은 먼저  $\mathbb{R}^{\times}$  구조 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

12-25. Fall

10m

5. Let  $G$  be an order 21 subgroup of  $\mathbb{C}^\times$ .

a)  $G$  is cyclic.

b). For any  $g \in G$ ,  $x^{12} = g^{15}$  has 3 distinct solutions in  $G$ .

Proof. a). Since  $\mathbb{C}$  is abelian, so  $G$  is also abelian.

By the Cauchy's theorem,  $G$  has order 3 element  $x$  and order 7 element  $y$ .  
Now, since 3 and 7 are relatively prime,  $|xy| = 21$ , so  $G = \langle xy \rangle$ .

b). By a),  $G \cong \mathbb{Z}/21\mathbb{Z}$ . So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of  $g$ ,  $\gcd(12, 21) = 3$  divides  $15g$ , so it has three distinct solutions.

Note:  $ax \equiv b \pmod{n}$  has a solution iff  $\gcd(a, n) \mid b$ .

And in such a case, it has  $\gcd(a, n)$  solutions.

2025. Fall.

6. Let  $\{a_n\}$  be a bounded sequence in  $\mathbb{R}$ . Define

22m

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that  $\{a_n\}^*$  is closed.

b). Show  $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence  $\{b_n\}$  such that  $\{b_n\}^* = [0, 1]$ ?

Proof a). If  $\{a_n\}^*$  is finite, then clearly closed in  $\mathbb{R}$ .

Assume that  $\{a_n\}^*$  is infinite. Since  $\{a_n\}$  is bounded, so the set of subsequential limits  $\{a_n\}^*$  is bounded set. So it has a limit point.

Let  $\alpha$  be a limit point of  $\{a_n\}^*$ . Then, given  $\epsilon \in \mathbb{R}$ , there exists a subsequential limit  $p$  of  $\{a_n\}$  such that  $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ .

For  $k=1$ , choose such a  $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ , and since  $p$  is a subsequential limit, choose  $a_{p_1} \in B_{\frac{1}{2}}(p)$ , then  $a_{p_1} \in B_1(\alpha)$ .

For chosen  $a_{p_1}, \dots, a_{p_{k-1}}$ , choose  $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ .

Then, similarly, there is  $a_{p_k} \in B_{\frac{1}{2}}(p_k)$  such that  $p_k > p_{k-1}$ .

because no such  $p_k$  exists, then it is contradiction to  $a_{p_k}$  is a subsequential limit. Therefore, the arbitrary limit point  $\alpha$  is contained in  $\{a_n\}^*$ .

so it is closed.

b). Since a),  $\{a_n\}^*$  is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  such that  $n \geq N \Rightarrow a_n < \alpha + \epsilon$  (since  $\alpha = \max \{a_n\}^*$ ) and there exists infinitely many terms in  $\{a_n\}$  such that  $\alpha - \epsilon < a_n$  (since  $a_{p_k} \rightarrow \alpha$ ).

Thus  $\limsup_{n \rightarrow \infty} a_n = \alpha$ .

c). Let  $b_n$  be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given  $b \in [0, 1]$ , there exists a subsequence in  $\{b_n\}$  converges to  $b$ , because: for any  $\epsilon > 0$ , there are infinitely many terms in  $b_n$  such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let  $E$  be a nonempty bounded subset of a Metric space  $(X, d)$ .

10m

Define  $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove  $\text{diam } E = \text{diam } \bar{E}$

b). If  $E$  is compact, then  $d(x, y) = \text{diam } E$  for some  $x, y \in E$ .

Proof.

a). Since  $E \subseteq \bar{E}$ ,  $\text{diam } E \leq \text{diam } \bar{E}$  is clear.

To show the inverse inequality, using the property of supremum.

Given  $\varepsilon > 0$ , there exists  $x, y \in \bar{E}$  such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since  $\bar{E}$  is closure, there are  $x', y' \in E$  such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now,  $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$ .

Since  $\varepsilon > 0$  was arbitrary,  $\text{diam } \bar{E} \leq \text{diam } E$ , so proved. 7m.

b). Since  $E$  is compact set, so  $E \times E$  is also compact set.

Since the metric function  $d$  is continuous, it preserves compactness, so  $d[E \times E] \subseteq \mathbb{R}$  is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is  $(x', y') \in E \times E$  such that  $d(x', y') = \max d[E \times E]$  3m.

2025 Fall.

8. Let  $a > 1$ . Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore  $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

Now, 
$$\int_t^{at} \frac{\cos x}{x} dx = \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx$$
$$= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx$$

(It is possible, because  $\sum (-1)^n x^{2n-1}/(2n)!$  converges for all  $x \in \mathbb{R}$ )

Now, 
$$\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0,$$
 so

the given limit is  $\ln a$ .

Revised in next

Observation for  $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$ .

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all  $x \neq 0$ . Therefore, for any  $t > 0$ ,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \Big|_t^a \end{aligned}$$

Meanwhile, the series converges uniformly in  $[t, a]$  for small enough  $t > 0$ , specifically,  $t < \frac{1}{a}$ , then for each  $n \in \mathbb{N}$ ,

$$\left| (-1)^n \frac{x^{2n}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left( \frac{1}{2n \cdot (2n)!} \cdot (a^{2n} t^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note:  $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$ ,  $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} \rightarrow 0$  as  $t \rightarrow 0$ .

2025. Fall.

12. Define an equi. rel on  $\mathbb{R}$  as  $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$ .

" as  $x \sim y \Leftrightarrow x = y$  or  $x - y \in \mathbb{Q}$ .

a). Describe the quotient topology on  $\mathbb{R}/\sim$ .

b). " on  $\mathbb{R}/\sim$ .

c). In  $\mathbb{R}/\sim$ , determine if every point is closed.

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since  $\mathbb{Q}$  is closed under the operation  $-$ , so these relations are equi. rel.

a). First,  $\mathbb{R}/\sim = \{a + \mathbb{Q} \mid a \in \mathbb{R}\}$ , where  $a + \mathbb{Q} = \{a + q \mid q \in \mathbb{Q}\}$ . And, given a basis element  $(a, b) \in \mathbb{R}$ , for the projection  $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim: a \mapsto a + \mathbb{Q}$ .

$$\pi^{-1}[\pi[(a, b)]] = \bigcup_{r \in (a, b)} r + \mathbb{Q} = \bigcup_{q \in \mathbb{Q}} (a + q, b + q)$$

is open and since  $\pi$  is quotient map, so  $\pi[(a, b)]$  is open, so  $\pi$  is open. Therefore, the topology on  $\mathbb{R}/\sim$  is generated by

$$\mathcal{B}_{\sim} = \{\pi[(a, b)] \mid a < b\} = \{(a, b) + \mathbb{Q} \mid a < b\}.$$

But, by the density of rationals, for any  $a < b$ ,  $(a, b) + \mathbb{Q} = \mathbb{R}/\mathbb{Q}$ . Therefore the topology on  $\mathbb{R}/\sim$  is indiscrete.



2029, Fall.

13. Consider a regular plane curve  $C: I \rightarrow \mathbb{R}^2$ .

a). Give a formula for the normal  $n$  to  $C$ , and define the signed curvature.

b). Explain the relation between the total angle change and the signed curvature.

c). Assume  $C$  is a closed plane curve parametrized by arc-length, and  $I = [0, L]$ .

Prove or disprove that: there are  $t, t' \in I$  such that  $C'(t) = -C'(t')$ .

and if  $t=0$ , there is such a  $t' \in I$ ?

Moreover, show that  $\int_0^L |k(s)| ds \geq 2\pi$  where  $k(s)$  is the signed curvature.

Proof.

TSNU. Q. 16.

Prove that  $\lim_{b \rightarrow \infty} \int_0^b \frac{\sin x}{x} dx$  exists.

Proof. First, let  $a_n = n\pi$  ( $n \in \mathbb{N}$ ). Then, a sequence  $\left\{ \int_0^{a_n} \frac{\sin x}{x} dx \right\}$  converges, because:

$\left| \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx \right|$  is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx$  is positive if  $n$  is even and is negative if  $n$  is odd.

That is,  $\int_0^{n\pi} \frac{\sin x}{x} dx = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} \frac{\sin x}{x} dx$  is alternating series, so converges.

7226. Prima.

3. Let  $V$  be an  $n$ -dimensional vector space over  $\mathbb{C}$ ,  
Let  $T: V \rightarrow V$  be a linear operator such that  $T^3 = T$ .

27m

a). Find all possible characteristic polynomials and corresponding minimal polynomials for  $T$ .

b). Prove  $V = \ker T \oplus \operatorname{Im} T$ .

Proof.

a). Since  $T^3 - T = 0$ , the minimal polynomial  $m(t)$  for  $T$  divides  $t^3 - t = 0$ .  
 $t^3 - t = t(t-1)(t+1)$ , there are possibilities:

$$m(t) = t \text{ or } t-1 \text{ or } t+1 \text{ or } t(t-1) \text{ or } t(t+1) \text{ or } (t-1)(t+1) \text{ or } t(t-1)(t+1).$$

Since  $m(t)$  divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3} \text{ where } r_1, r_2, r_3 \text{ are non-negative integers and } r_1 + r_2 + r_3 = n.$$

If  $r_1 = 0$  and  $r_2, r_3 \neq 0$ , then the corresponding minimal polynomial is  
 $(t-1)(t+1)$

and other cases are similar.

b). ~~Given  $\alpha \in \ker T \cap \operatorname{Im} T$ ,  $T\alpha = 0$  and  $\alpha = T\beta$  for some  $\beta \in V$ .~~

Given  $\alpha \in V$ ,  $\alpha = T^2\alpha + \alpha - T^2\alpha$  and since  $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$ ,  
so  $\alpha - T^2\alpha \in \ker T$  and trivially  $T^2\alpha \in \operatorname{Im} T$ . Moreover if  $T\alpha = 0$  and  $\alpha \in \operatorname{Im} T$ ,  
Then  $\alpha = T\beta = T^3\beta$  for some  $\beta \in V$  and  $T\alpha = T^2\beta = 0$  so  $T^3\beta = TT^2\beta = \alpha = 0$ ,  
so  $V = \ker T \oplus \operatorname{Im} T$ .

Modified.

Let  $T$  be a linear operator on the fin. dim space  $V$  over  $\mathbb{C}$  satisfying  $T^4 = T^2$ .

a). Prove that  $V = \ker T^2 \oplus \operatorname{Im} T^2$ .

b). Describe  $T|_{\operatorname{Im} T^2}$ .

c). Prove that  $T$  is diagonalizable  $\Leftrightarrow \ker T = \ker T^2$ .

observe:  $T^4 - T^2 = 0$ , so the minimal polynomial divides  $t^2(t-1)(t+1)$ .

a). Given  $\alpha \in V$ ,  $\alpha = T^2\alpha + \alpha - T^2\alpha$  and  $T^2\alpha \in \operatorname{Im} T^2$  clearly, and  $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$ , so  $\alpha - T^2\alpha$ , which proves  $\alpha - T^2\alpha \in \ker T^2$ , so  $V = \ker T^2 + \operatorname{Im} T^2$ . Moreover, given  $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$ ,  $T^2\alpha = 0$  and  $\alpha = T^2\beta$  for some  $\beta \in V$ , so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

so  $V = \ker T^2 \oplus \operatorname{Im} T^2$ .

b). Given  $\alpha \in \operatorname{Im} T^2$ ,  $\alpha = T^2\beta$  for some  $\beta \in V$ . so,  $T\alpha = T^3\beta$ .

Multiplying  $T$ , then  $T^2\alpha = T^4\beta = T^2\beta$ , so  $T^2(\alpha - \beta) = 0$ .

Therefore,  $\alpha - \beta \in \ker T^2$ , but since  $\alpha \in \operatorname{Im} T^2$  and  $\operatorname{Im} T^2 \cap \ker T^2 = 0$ ,

$\beta = \alpha + \gamma$ , where  $\gamma \in \ker T^2$ . Now,  $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$ ,

so,  $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$ , moreover, since  $T^2 = T^4$ ,  $T^2|_{\operatorname{Im} T^2} = I$ .

c). Since  $T$  is diagonalizable  $\Leftrightarrow$  minimal polynomial of  $T$  splits and separable.

That is, the minimal polynomial is  $t^a(t-1)^b(t+1)^c$ ,  $a, b, c \in \{0, 1\}$ .

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4. Let  $G$  be a group and  $H_1, H_2 \leq G$  be subgroups of finite index.

a). Prove that the map

$$T: G \rightarrow S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and  $\text{Ker } T \subseteq H_1 \cap H_2$ .

b). Prove  $H_1 \cap H_2$  has finite index.

Proof

$$\begin{aligned} \text{a). Given } g_1, g_2 \in G, T(g_1 g_2)(aH_1, bH_2) &= (g_1 g_2 aH_1, g_1 g_2 bH_2) \\ &= T(g_1) \circ T(g_2)(aH_1, bH_2) \end{aligned}$$

for each  $a, b \in G$ , so it preserves the operation.

$$\text{And } g \in \text{Ker } T \Leftrightarrow T_g = \text{id} \Leftrightarrow \text{for all } a, b \in G, T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$$

$$\begin{aligned} &\Leftrightarrow \text{for all } a, b \in G, a^{-1}ga \in H_1 \text{ and } b^{-1}gb \in H_2 \\ &\Rightarrow g \in H_1 \cap H_2 \text{ (take } a=b=1_G) \end{aligned}$$

b). By the first-isomorphism theorem,

$$G/\text{Ker } T \cong \text{Im } T \leq S_{G/H_1 \times G/H_2}$$

and since  $H_1, H_2$  have finite index, so the Codomain of  $T$  is finite set.

$$\text{Now, } |G : H_1 \cap H_2| \leq |G : \text{Ker } T| \leq \# S_{G/H_1 \times G/H_2}$$

because  $\text{Ker } T \subseteq H_1 \cap H_2$ , so proved.

Note: If  $A \leq B \leq G$  and  $|G:A|$  is finite, then

$$G/A \rightarrow G/B : gA \mapsto gB$$

is well-defined and onto, because: Since  $A \leq B$ ,

$$g_1 A = g_2 A \Leftrightarrow g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \Leftrightarrow g_1 B = g_2 B$$

and surjectiveness is clear.

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8. For  $f \in C([0,1])$ , define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x < 0 \\ f(x) & \text{if } 0 \leq x \leq 1 \\ f(1) & \text{if } x > 1 \end{cases}$$

For each  $n \geq 1$ , define

$$p_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define  $T_n f(x) := \int_{\mathbb{R}} p_n(x-y) \tilde{f}(y) dy$ ,  $x \in \mathbb{R}$ .

a). Show that  $T_n f \rightarrow f$  on  $[0,1]$ .

b). Determine  $T_n f \rightarrow f$  uniformly on  $[0,1]$  or not.

Proof. Given  $x \in \mathbb{R}$  and  $n \in \mathbb{N}$ ,  $x-y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n}+x \leq y \leq \frac{1}{n}+x$ . Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y) dy.$$

Since  $\tilde{f}$  is continuous on  $[-1,2]$ , so it is continuously uniformly.

Given  $\varepsilon > 0$ , take  $\delta > 0$  such that  $\forall x, t \in [0,1]$ ,  $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$ .

Now, given  $x \in [0,1]$ , and  $n > \frac{1}{\delta}$ ,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y) dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} f(x) dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} [\tilde{f}(y) - f(x)] dy \right| \\ &\leq \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)| dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore,  $\sup_{x \in [0,1]} |T_n f - f| \rightarrow 0$  as  $n \rightarrow \infty$ , so converges uniformly.

